Strength and finish knowledge organiser

Design and Technology LAUNCH Spring 2 Week 4 Thursday 18 March 2027 P1 Year A

Weekly outcome verbatim: CONSTRUCTION: evaluate strength and finish.

I can check my actual stand, make a useful change and explain the result.

Today I can

[ ] I can use the agreed one-card test safely and record what happened.

[ ] I can check strength and finish as different features.

[ ] I can make one approved improvement and log why it helps, then recheck.

Feature	Meaning
Strength	Do the body and joins support the intended one-card load?
Finish	Are edges, folds, tape and channel neatly made and safe?

Our intended load

One 80 mm by 60 mm lightweight paper card. Level table. Central card position. Hands away for 30 seconds after adult clearance. Stop at the first concern. The whole stand fits a 160 mm by 120 mm table area; body base alone is 120 mm by 80 mm.

A fair change check

Keep card, position, table and timing target the same before and after. Record actual count, actual seconds and real observations. Add the improvement, why it helps, support and next step.

Resilience and change

A stopped test gives useful information. Improve safely and explain the evidence. Missing practical work stays pending. More cards are not required.

In your book. Plan in any language. Point, speak, sign or confirm a scribe's words.

The real stand and its useful job

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Invented client Sam needs one lightweight paper prompt card, 80 mm wide and 60 mm high, upright and readable on a level table. The whole stand, including its channel, should fit a 160 mm wide by 120 mm deep table area. It should hold the card for 30 seconds with hands away. The card should lift out by hand. Plan safe smooth edges and no loose sharp parts. These are invented design targets, not standards.

Retrieve the actual work

Retrieve the actual approved design, build plan, Week 2 mock-up or Week 3 made item and change log. The default triangular body base is 120 mm by 80 mm; the U channel projects beyond the front. Use the actual approved revised plan if it differs. A missing or unfinished item stays honestly marked. Do not relabel an adult model as pupil work.

What we evaluate today

Strength: how the body and joins support the intended one-card load. Finish: the quality of edges, folds, tape and channel. A neat surface alone does not show strength. A standing card alone does not show that every edge is safe.

The normal use load test

The adult clears the actual item and level table. Use one actual 80 mm by 60 mm lightweight paper card, centrally in the U channel against the slope. Hands away for 30 seconds. Stop at the first concern. Remove the card and inspect. No heavy objects or force are needed.

Improve and keep

Use one real finding to agree a useful safe improvement. Make it with the approved support, then repeat the same one-card check. Keep actual count, time, result and support. If an action cannot happen, mark it pending.

Normal use load test card

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Use the intended one-card load first. A normal-use test checks the job the stand is meant to do. Stop at the first concern. Do not test to failure.

1 Clear and inspect

Put tools down and return them as agreed. The adult checks actual edges, folds, joins, channel, table and safe support. Damaged or suspect edges are handled by the adult. Any actual approved adhesive join must complete the full cure on its checked label before testing.

2 Set up the same test

Use a clear level table. Place the stand in a 160 mm by 120 mm overall table area. Use one actual lightweight paper card 80 mm wide by 60 mm high. Put it centrally in the U channel against the slope.

3 Hands away for 30 seconds

After adult clearance, keep hands away and observe for 30 seconds. Record the actual count, time and result. Stop at the first slip, tilt, bow, loose join, damage, concern or adult stop signal.

4 Remove and inspect

Gently remove the card. Look at folds, joins and channel with the adult. Do not touch suspect edges. A planned test is not a result.

5 Improve and recheck

Agree one safe useful change linked to a real finding. Make it with approved support. Repeat the one-card test with the same card, position, table and 30-second target after adult clearance. Log the actual before and after. A new adhesive repair keeps the recheck pending until its full checked label cure.

Stop signs

Slip. Tilt. Bowing. Loose join. Damage. Concern. Adult stop signal. Stop at the first sign; follow the adult. No one holds the stand steady during the hands-away test. Do not add another card after a stop.

Use the actual approved revised plan if it differs. One-card purpose stays clear. A model demonstration is not a pupil result.

Before and after load test record

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In your book or below, record only what was observed. A planned 30 seconds is not an actual timing. Keep the same card, position, level table and timing target for the fair one-card comparison.

Record	Before improvement	After improvement
Actual card count	___	___
Actual time in seconds	___	___
Supported upright or stop sign	___	___
Join fold and channel finding	___	___
Actual role and support	___	___
Actual date and location	___	___

A stopped or missing test

Test stopped / not performed ___. Actual reason ___. Adult next step ___. Keep the result pending if it did not happen. Never copy a target as an actual result.

Optional Stretch comparison

Optional Stretch comparison only: first pass the one-card baseline and ask the adult to clear the next stage. Use a second identical lightweight paper card, then a third, maximum three cards total. Each is 80 mm by 60 mm and made from the same lightweight paper; no added clips, lamination or weights. Observe for 30 seconds at each stage. Remove cards and inspect between stages. Stop at the first concern; never add another card after a stop. Keep card size and material, position, level table and timing the same. The purpose is still one card. This is a comparison, not a maximum load or safety claim.

Record each actual comparison stage separately: card count ___. Actual time ___. What happened ___. Cards removed and inspected ___. Adult clearance ___. No extra-card result is a maximum capacity or safety certificate.

Finish quality check

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Inspect only after the adult clears the work. Put tools down first. Do not touch suspect edges. Use the list on the actual stand; a model picture is not a tactile check.

Finish feature	Actual finding and next step
Safe smooth edges; ask the adult about any suspect edge.	_____
Neat folds that keep the planned shape.	_____
Paper tape lies flat and secure, without loose ends.	_____
U channel is aligned and open, with its front stop up toward the card.	_____
No loose bits or sharp parts.	_____
Card message stays readable and the card lifts out by hand.	_____

Keep the two checks clear

Strength is support under the intended load. Finish is the quality of edges, folds and joins. The two can affect each other: a loose tape end is a finish finding, and a loose joint may also affect strength.

Choose a useful improvement

A finish finding was ___. The adult approved ___. It helps because ___. My actual action was ___. Any adult action was ___. A check still pending is ___.

Do not add decoration that hides the card message or blocks the channel. Do not make a fault to create a change. If no safe useful change is suitable, agree a next step and keep it pending.

My improvement and recheck log

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Keep your own words. One real approved improvement needs a finding and a reason. Do not invent a fault, change or successful test.

Before

My actual item or model was ___. The one-card test showed ___ / was stopped or not performed because ___. The finish check showed ___. My evidence location was ___.

The agreed change

I improved ___ by ___. The adult approved it because ___. This helps Sam's purpose or safe handling by ___. My action was ___. The adult did ___. Actual date ___.

After

I used the same one-card check: same card, same central position, same level table and 30-second target. Actual count ___. Actual seconds ___. What really happened ___. Finish finding after the change ___. Adult clearance and support ___.

My evaluation

The improvement helped / needs more work because ___. I know this from ___. One limit or next step is ___. Scribed words confirmed yes / not yet ___.

If the action is pending

My proposed improvement is ___. The reason is ___. Actual making or recheck remains pending because ___. Keep proposal and action separate.

You may record more real adjustments on another page. Return to the one-card check after each approved change. A higher card count is never required for the core outcome.

Strength test and repair safety brief

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Follow the actual adult-approved route. All tests are gentle tabletop checks with lightweight paper cards. A quiz answer is not permission to use tools or add load.

1 Use the intended one lightweight paper card first. A normal-use load test does not need a heavy weight or force.

2 The adult clears actual joins, edges, tool return, table and support before the test. Keep hands away while observing; no one steadies the stand.

3 Stop for slip, tilt, bowing, loose join, damage, concern or the adult signal. Remove the load safely as directed. Let the adult handle damaged or suspect parts.

4 No pushing, shaking, dropping, heavy objects or tests to failure. No hot glue, heated tools or power tools are used.

5 Any repair uses only the actual approved hand-tool method and support. Close scissors and put them down before speaking or pausing. Ask before a new cut or join.

6 After a change, return to one card first. Record a stopped or unperformed test honestly. Do not add another card to see what happens after a warning.

7 An actual approved adhesive join must complete its full label cure before testing. The adult checks the actual instructions; never guess a cure time. A new adhesive repair keeps the recheck pending until full cure. The default uses cold paper tape.

Safe stop routine

Tools down. Stop the test at the first concern. Follow the adult to remove the load and inspect. Agree one safe improvement. Recheck with one card first after clearance. Resume only when the adult confirms the route.

No item or test space

With no actual stand or safe test space, use the model brief and spoken checks. Point, speak or sign a strength check, a finish check and a proposed improvement with a reason. Rehearse the stop routine with empty hands. An adult may scribe exact confirmed words. Label this model evaluation or planning. Actual load testing, physical improvement and the recheck stay pending unless performed.